22. $a = 48.8$; $b = 2$; $Y = 48.8 + 2X$

When $X = 1992$, $\hat{Y} = 48.8 + 2\ (1992\text{-}1994) = 44.8$

When $X = 1993$, $\hat{Y} = 48.8 + 2\ (1993\text{-}1994) = 46.8$

When $X = 1994$, $\hat{Y} = 48.8 + 2\ (1994\text{-}1994) = 48.8$

When $X = 1995$, $\hat{Y} = 48.8 + 2\ (1995\text{-}1994) = 50.8$

When $X = 1996$, $\hat{Y} = 48.8 + 2\ (1996\text{-}1994) = 52.8$

When $X = 1997$, $\hat{Y} = 48.8 + 2\ (1997\text{-}1994) = 54.8$

Exercise 9.2

14. Laspeyre's IN = 144.8 Paasche's IN = 144.4
15. Laspeyre's IN = 164.5 Paasche's IN = 162.4
16. Laspeyre's IN = 106.6 Paasche's IN = 106.8 Fisher's IN= 106.7
17. Fisher's IN = 138.5 TRT= 1 FRT = 1880/1560
18. Fisher's IN = 83.6
19. Fisher's IN = 122.314 TRT = 1
20. Cost of Living Index =2662.38
21. Cost of Living Index = 117.31
22. Cost of Living Index = 130.6192

Exercise 9.3

14. $\overline{\overline{X}} = 16.2$, UCL = 20.49, CL = 16.2, LCL = 11.91

    $\overline{R} = 7.4$, UCL = 15.65, CL = 7.4, LCL = 0
15. $\overline{\overline{X}} = 46.2$, UCL = 50.14, CL = 46.2, LCL = 42.26

    $\overline{R} = 6.8$, UCL = 14.38, CL = 6.8, LCL = 0
16. $\overline{\overline{X}} = 37.7$, UCL = 48.14, CL = 37.7, LCL = 27.26

    $\overline{R} = 18$, UCL = 38.07, CL = 18, LCL = 0
17. $\overline{\overline{X}} = 10.66$, UCL = 14.31, CL = 10.66, LCL = 7.006

    $\overline{R} = 6.3$, UCL = 13.32, CL = 6.3, LCL = 0
18. $\overline{\overline{X}} = 12.5$, UCL = 12.71, CL = 12.5, LCL = 12.28

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